

1 Summary

Model (Run-name): model3 (test)

Iteration: 3

TAC: 60749.079874196184

Took: 11.01643681526184s

Total Time: 22.246309280395508s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

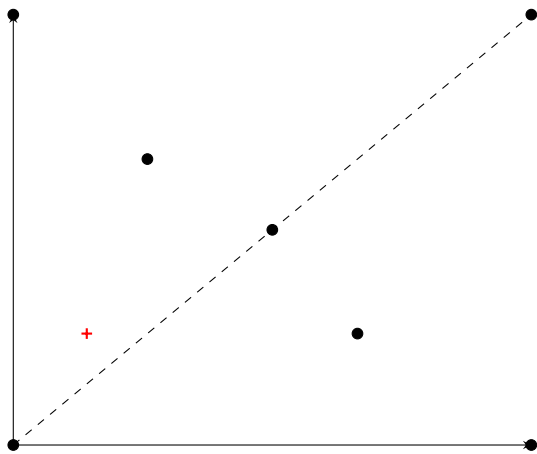
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{3,5,2}$	22.3	$q_{1,1,2}$	586.9599999999999
$z_{2,2,1}$	1.0	$dt_{3,5,3}$	29.056081081081047	$q_{2,2,1}$	647.2099999999999
$z_{2,4,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{2,4,2}$	530.45
$z_{3,5,2}$	1.0	$dt_{4,3,3}$	111.1	$q_{3,5,2}$	1544.2900000000002
$z_{4,3,2}$	1.0	$dt_{5,1,1}$	38.9	$q_{4,3,2}$	1539.7200000000003
$z_{5,1,1}$	1.0	$dt_{5,1,2}$	14.404360127374531	$q_{5,1,1}$	173.03999999999996
$z_{5,4,2}$	1.0	$dt_{5,4,2}$	12.423728813559336	$q_{5,4,2}$	1104.4
$z_{cu,3}$	1.0	$dt_{5,4,3}$	44.5282485875706	$q_{cu,3}$	839.9899999999994
$z_{cu,5}$	1.0	$dt_{cu,3}$	82.35608108108104	$q_{cu,5}$	1081.9699999999998
$A_{1,1,2}^\beta$	4.659468026433878	$dt_{cu,5}$	86.7282485875706	$t_{1,1}^{(H)}$	160.0
$A_{2,2,1}^\beta$	4.284212301867288	$f_{1,1,2}^H$	8.8	$t_{1,2}^{(H)}$	160.0
$A_{2,4,2}^\beta$	0.5207742819827613	$f_{2,2,1}^H$	5.8254725472547255	$t_{1,3}^{(H)}$	93.3
$A_{3,5,2}^\beta$	8.867082014975438	$f_{2,4,2}^H$	4.774527452745273	$t_{2,1}^{(H)}$	248.9
$A_{4,3,2}^\beta$	6.406119630943358	$f_{3,5,2}^H$	14.8	$t_{2,2}^{(H)}$	187.84245283018686
$A_{5,1,1}^\beta$	0.24292234696382392	$f_{4,3,2}^H$	12.6	$t_{2,3}^{(H)}$	137.8
$A_{5,4,2}^\beta$	2.30215155049158	$f_{5,1,1}^H$	17.7	$t_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	4.83739822624145	$f_{5,4,2}^H$	8.285071267816955	$t_{3,2}^{(H)}$	226.7
$A_{cu,5}^\beta$	5.7530214433620355	$f_{1,1,2}^C$	5.869	$t_{3,3}^{(H)}$	122.35608108108104
$A_{1,1,2}$	20.05363555217427	$f_{2,2,1}^C$	6.1	$t_{4,1}^{(H)}$	271.1
$A_{2,2,1}$	13.879417919656465	$f_{2,4,2}^C$	5.613227513227513	$t_{4,2}^{(H)}$	271.1
$A_{2,4,2}$	3.743604220332404	$f_{3,5,2}^C$	13.9	$t_{4,3}^{(H)}$	148.9
$A_{3,5,2}$	55.82566076118323	$f_{4,3,2}^C$	8.4	$t_{5,1}^{(H)}$	198.9
$A_{4,3,2}$	23.668855770059224	$f_{5,1,1}^C$	2.4792233009708715	$t_{5,2}^{(H)}$	189.12372881355932
$A_{5,1,1}$	1.4656333375682882	$f_{5,4,2}^C$	11.686772486772485	$t_{5,3}^{(H)}$	126.7282485875706
$A_{5,4,2}$	11.133625686778567	$RecLMTD_{1,1,2}$	0.029040463096885894	$t_{1,1}^{(C)}$	160.0
$A_{cu,3}$	16.738997089244645	$RecLMTD_{2,2,1}$	0.0182282493034841	$t_{1,2}^{(C)}$	137.23157894736843
$A_{cu,5}$	20.944873037188852	$RecLMTD_{2,4,2}$	0.026432601068855872	$t_{1,3}^{(C)}$	60.0
$dt_{1,1,2}$	22.768421052631595	$RecLMTD_{3,5,2}$	0.030727267318318277	$t_{2,1}^{(C)}$	221.7
$dt_{1,1,3}$	33.3	$RecLMTD_{4,3,2}$	0.013066354534948133	$t_{2,2}^{(C)}$	115.6
$dt_{2,2,1}$	27.2	$RecLMTD_{5,1,1}$	0.027943700892763483	$t_{2,3}^{(C)}$	115.6
$dt_{2,2,2}$	72.24245283018718	$RecLMTD_{5,4,2}$	0.03421665963078272	$t_{3,1}^{(C)}$	221.1
$dt_{2,4,2}$	11.142452830186798	$RecLMTD_{cu,3}$	0.016938472512598907	$t_{3,2}^{(C)}$	221.1
$dt_{2,4,3}$	55.6	$RecLMTD_{cu,5}$	0.016454376814154296	$t_{3,3}^{(C)}$	37.8

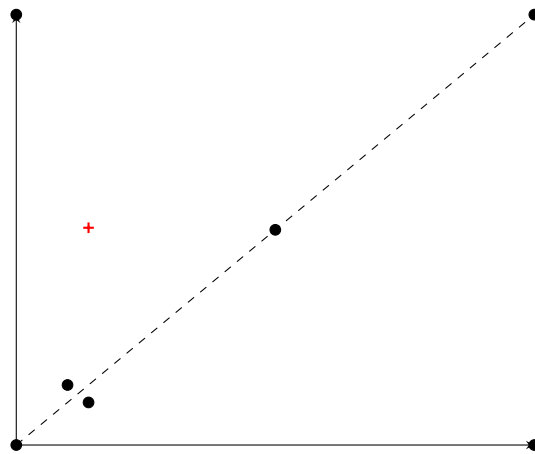
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,2}^{H,out}$	543.5006751687922
$t_{4,2}^{(C)}$	176.7	$b_{1,1,2}^{C,in}$	352.176
$t_{4,3}^{(C)}$	82.2	$b_{2,2,1}^{C,in}$	705.16
$t_{5,1}^{(C)}$	204.4	$b_{2,4,2}^{C,in}$	461.4073015873016
$t_{5,2}^{(C)}$	204.4	$b_{3,5,2}^{C,in}$	1296.87
$t_{5,3}^{(C)}$	93.3	$b_{4,3,2}^{C,in}$	317.52
$t_{1,1,2}^H$	93.3	$b_{5,1,1}^{C,in}$	223.6357281553395
$t_{2,2,1}^H$	137.8	$b_{5,4,2}^{C,in}$	960.6526984126983
$t_{2,4,2}^H$	137.8	$b_{1,1,2}^{C,out}$	939.136
$t_{3,5,2}^H$	122.35608108108104	$b_{2,2,1}^{C,out}$	1352.37
$t_{4,3,2}^H$	148.9	$b_{2,4,2}^{C,out}$	991.8573015873015
$t_{5,1,1}^H$	189.12372881355932	$b_{3,5,2}^{C,out}$	2841.1600000000003
$t_{5,4,2}^H$	65.6	$b_{4,3,2}^{C,out}$	1857.2400000000002
$t_{1,1,2}^C$	144.10854085872572	$b_{5,1,1}^{C,out}$	396.67572815533947
$t_{2,2,1}^C$	221.7	$b_{5,4,2}^{C,out}$	2065.052698412698
$t_{2,4,2}^C$	112.86184971098265		
$t_{3,5,2}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,1,1}^C$	92.62135922330096		
$t_{5,4,2}^C$	146.03815028901732		
$b_{1,1,2}^{H,in}$	1408.0		
$b_{2,2,1}^{H,in}$	1449.9601170117012		
$b_{2,4,2}^{H,in}$	1188.3798829882985		
$b_{3,5,2}^{H,in}$	3355.16		
$b_{4,3,2}^{H,in}$	3415.8599999999997		
$b_{5,1,1}^{H,in}$	3520.5300000000007		
$b_{5,4,2}^{H,in}$	1647.9006751687918		
$b_{1,1,2}^{H,out}$	821.0399999999998		
$b_{2,2,1}^{H,out}$	802.7501170117014		
$b_{2,4,2}^{H,out}$	657.9298829882987		
$b_{3,5,2}^{H,out}$	1810.8699999999994		
$b_{4,3,2}^{H,out}$	1876.1399999999996		
$b_{5,1,1}^{H,out}$	3347.4900000000007		

3 *RecLMTD* Tangent Points

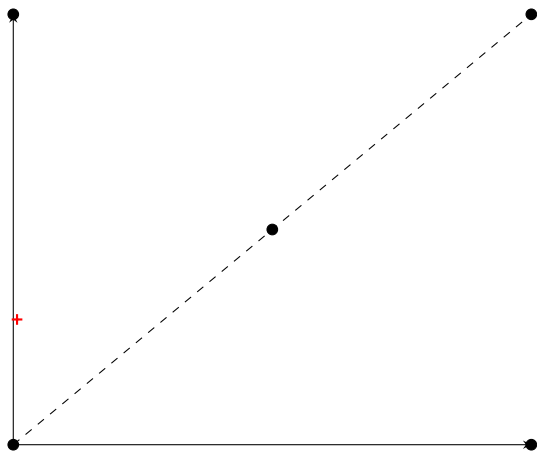
- (1, 1, 2)
 - Value: 0.029040
 - Error: 0.007058859192188
 - Relative error: 0.195539936613287
- (2, 2, 1)
 - Value: 0.018228
 - Error: 0.003458200429155
 - Relative error: 0.159463649965254
- (2, 4, 2)
 - Value: 0.026433
 - Error: 0.009723707838857
 - Relative error: 0.268935301545190
- (3, 5, 2)
 - Value: 0.030727
 - Error: 0.008443534701918
 - Relative error: 0.215556850164978
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000874453881
 - Relative error: 0.000066919610990
- (5, 1, 1)
 - Value: 0.027944
 - Error: 0.012613039344100
 - Relative error: 0.310997364937030
- (5, 4, 2)
 - Value: 0.034217
 - Error: 0.005544580795719
 - Relative error: 0.139446876813814
- $cu, 3$
 - Value: 0.016938
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $cu, 5$
 - Value: 0.016454
 - Error: 0.000000000000000
 - Relative error: -0.000000000000000



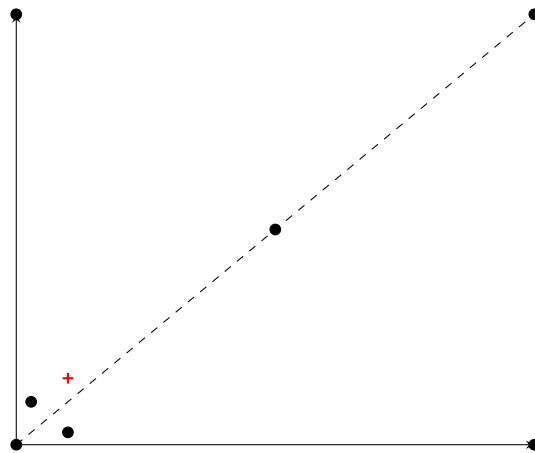
(a) *RecLMTD* - (1, 1, 2)



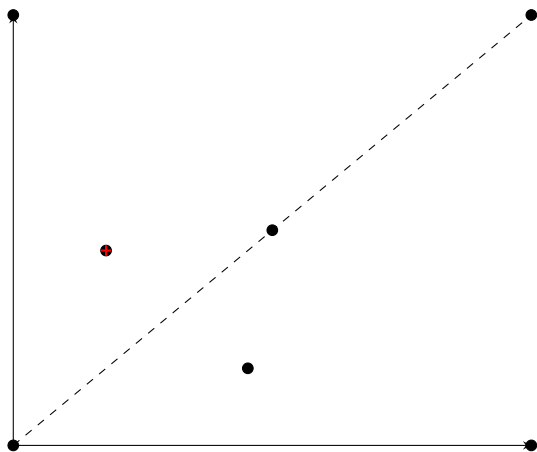
(b) *RecLMTD* - (2, 2, 1)



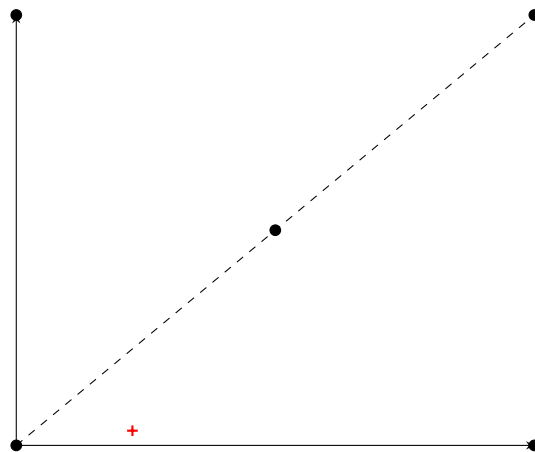
(c) *RecLMTD* - (2, 4, 2)



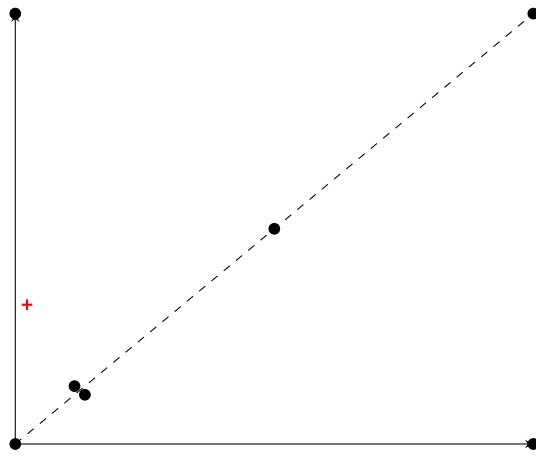
(d) *RecLMTD* - (3, 5, 2)



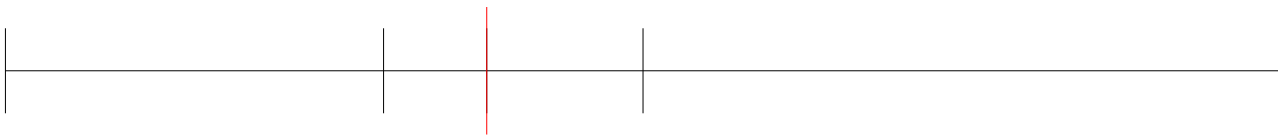
(e) *RecLMTD* - (4, 3, 2)



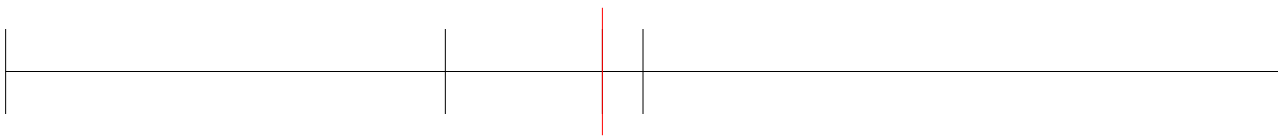
(f) *RecLMTD* - (5, 1, 1)



(g) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.9601170117012
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 1188.3798829882985
 - Absolute error: 291.5209351595623
 - Relative error: 0.3250465815893616
- (3, 5, 2)
 - Value: 3355.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 1, 1)
 - Value: 3520.53000000000007
 - Absolute error: 9.094947017729282e-13
 - Relative error: 2.58340278813965e-16
- (5, 4, 2)
 - Value: 1647.9006751687918
 - Absolute error: 80.99710351316594
 - Relative error: 0.05169246211340406

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 821.03999999999998
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -2.7693373702040345e-16
- (2, 2, 1)

- Value: 802.7501170117014
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.4162170183770757e-16
- (2, 4, 2)
 - Value: 657.9298829882987
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 1810.8699999999994
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 1, 1)
 - Value: 3347.4900000000007
 - Absolute error: 9.094947017729282e-13
 - Relative error: 2.7169452388892226e-16
- (5, 4, 2)
 - Value: 543.5006751687922
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,in}$

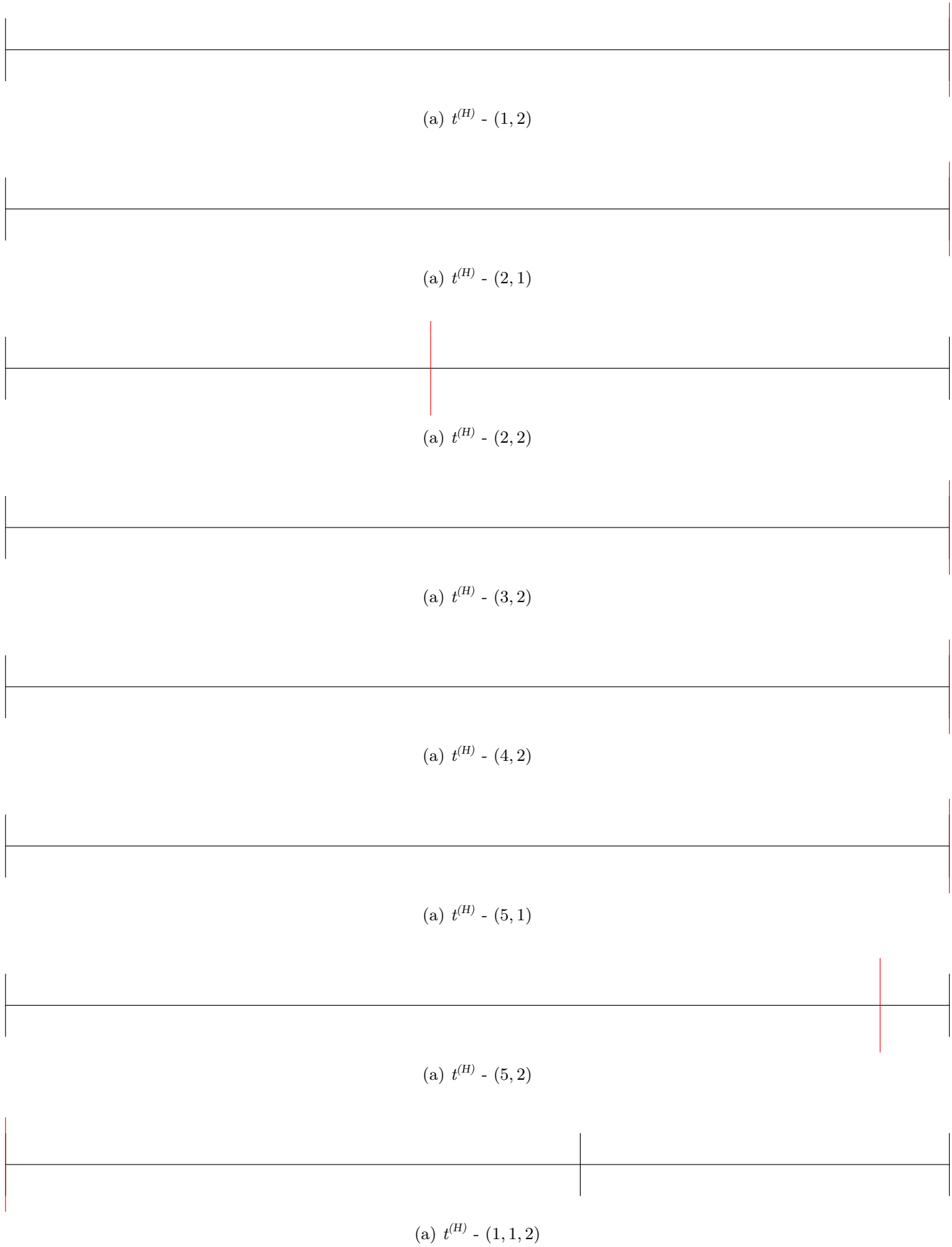
- (1, 1, 2)
 - Value: 352.176
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 461.4073015873016
 - Absolute error: 0.0
 - Relative error: 0.0

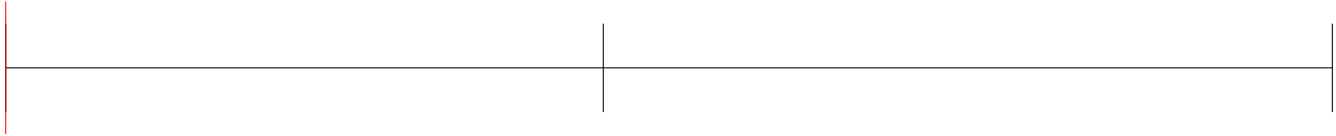
- (3, 5, 2)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 1, 1)
 - Value: 223.6357281553395
 - Absolute error: 116.59200000000004
 - Relative error: -0.342688118432155
- (5, 4, 2)
 - Value: 960.6526984126983
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

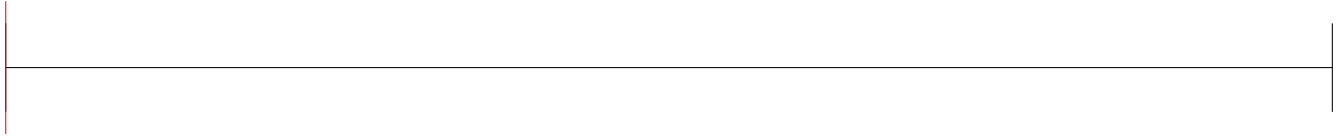
- (1, 1, 2)
 - Value: 939.136
 - Absolute error: 93.27650857562344
 - Relative error: 0.11027423528528528
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 991.8573015873015
 - Absolute error: 358.33806159586504
 - Relative error: 0.565630905859637
- (3, 5, 2)
 - Value: 2841.1600000000003
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.2400000000002
 - Absolute error: 2.2737367544323206e-13

- Relative error: 1.2242557528549464e-16
- (5, 1, 1)
 - Value: 396.67572815533947
 - Absolute error: 167.0466962013384
 - Relative error: 0.7274633123689731
- (5, 4, 2)
 - Value: 2065.052698412698
 - Absolute error: 358.3380615958654
 - Relative error: 0.20995780657520824





$$(a) \ t^{(H)} - (2, 2, 1)$$



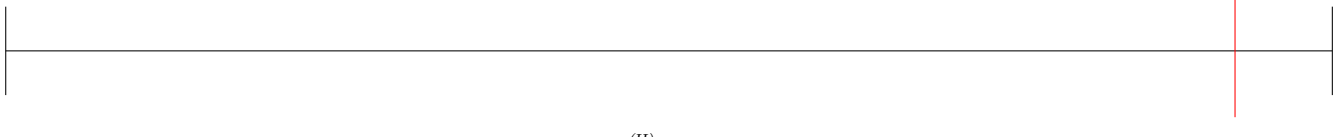
$$(a) \ t^{(H)} - (2, 4, 2)$$



$$(a) \ t^{(H)} - (3, 5, 2)$$



$$(a) \ t^{(H)} - (4, 3, 2)$$



$$(a) \ t^{(H)} - (5, 1, 1)$$



$$(a) \ t^{(H)} - (5, 4, 2)$$



$$(a) \ t^{(C)} - (1, 3)$$



$$(a) \ t^{(C)} - (2, 2)$$



$$(a) \ t^{(C)} - (4, 3)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (1, 2)$$



$$(a) \ t^{(C)} - (4, 3)$$



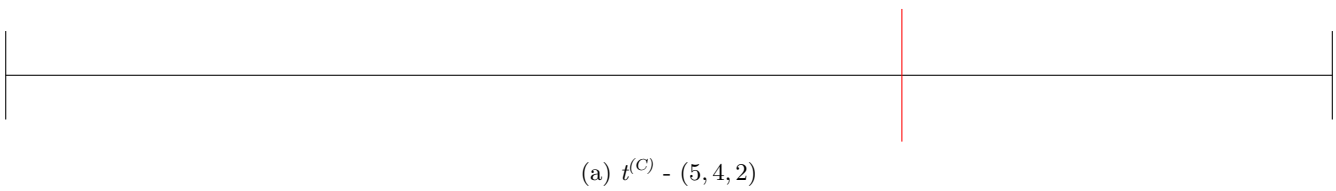
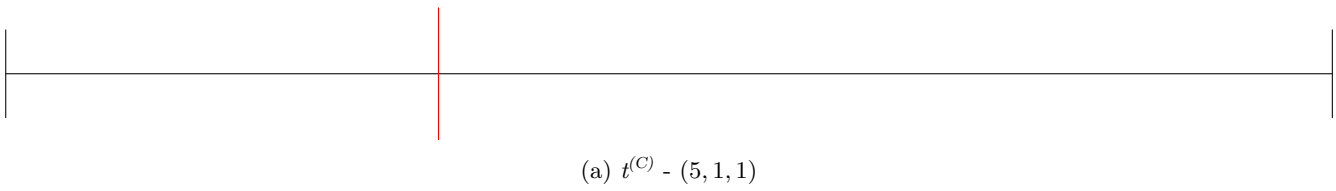
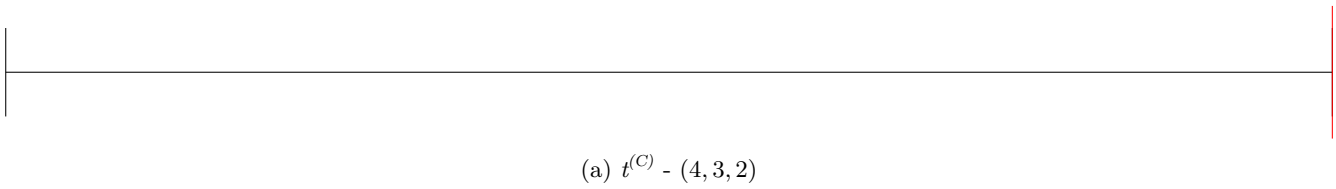
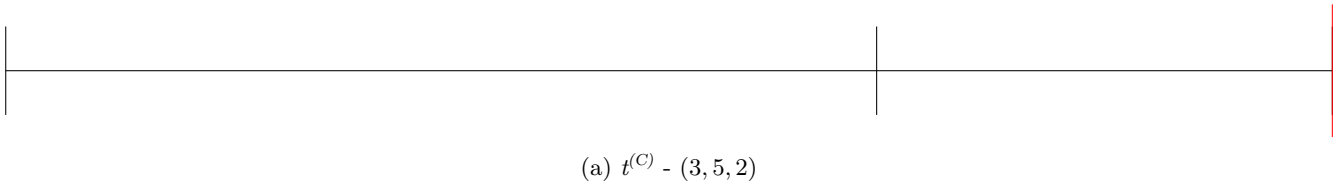
$$(a) \ t^{(C)} - (1, 1, 2)$$



$$(a) \ t^{(C)} - (2, 2, 1)$$



$$(a) \ t^{(C)} - (2, 4, 2)$$



5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 20.053636
 - Error: 0.000000000000014
 - Relative error: 0.000000000000001
- $(2, 2, 1)$
 - Value: 13.879418
 - Error: 0.000000000000060
 - Relative error: -0.000000000000004
- $(2, 4, 2)$
 - Value: 3.743604
 - Error: 12.751893705520065
 - Relative error: 0.773052972565001
- $(3, 5, 2)$
 - Value: 55.825661
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(4, 3, 2)$
 - Value: 23.668856
 - Error: 0.000000000000004
 - Relative error: 0.000000000000000
- $(5, 1, 1)$
 - Value: 1.465633
 - Error: 4.223046665353820
 - Relative error: 0.742359679782404
- $(5, 4, 2)$
 - Value: 11.133626
 - Error: 33.323878897029012
 - Relative error: 0.749567012566115

5.2 A_{CU}

- $cu, 3$
 - Value: 16.738997
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $cu, 5$

- Value: 20.944873
- Error: 0.0000000000000004
- Relative error: -0.0000000000000000



(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 4, 2)$



(a) $A^\beta - (3, 5, 2)$



(a) $A^\beta - (4, 3, 2)$



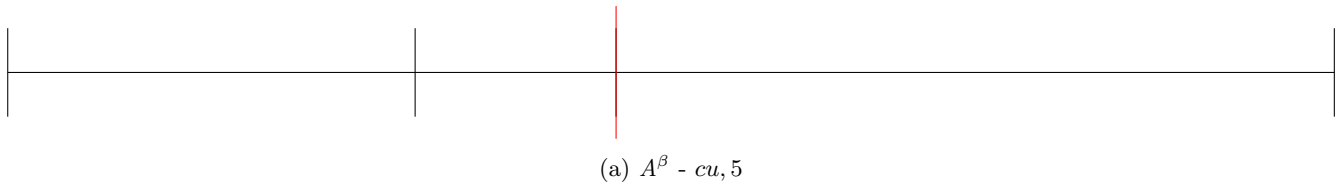
(a) $A^\beta - (5, 1, 1)$



(a) $A^\beta - (5, 4, 2)$



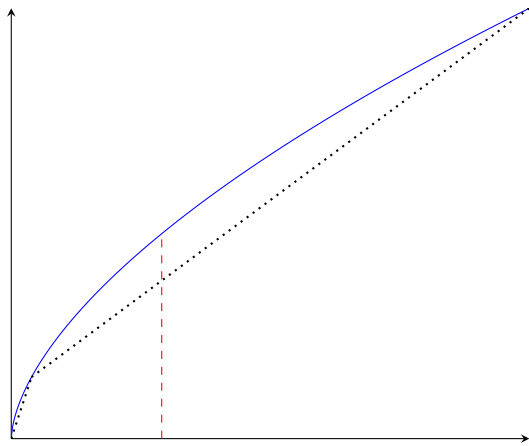
(a) $A^\beta - cu, 3$



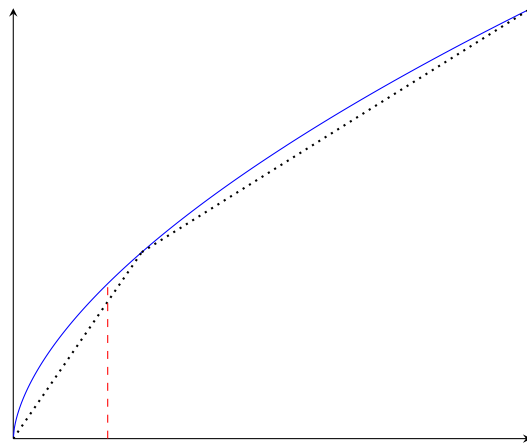
6 A^β Breakpoints

- (1, 1, 2)
 - q : 20.053636
 - $\hat{q}^\beta$: 4.659468
 - q^β : 6.043881
 - Breakpoints: [0, 2.7005882352941173, 69.05411764705883]
 - Error: 1.384412500324355
 - Relative error: 0.229060202992946
- (2, 2, 1)
 - q : 13.879418
 - $\hat{q}^\beta$: 4.284212
 - q^β : 4.846439
 - Breakpoints: [0, 18.890814020083003, 76.14235294117645]
 - Error: 0.562226708195442
 - Relative error: 0.116008208713260
- (2, 4, 2)
 - q : 3.743604
 - $\hat{q}^\beta$: 0.520774
 - q^β : 2.207872
 - Breakpoints: [0, 138.54823529411763]
 - Error: 1.687098048906210
 - Relative error: 0.764128444069464
- (3, 5, 2)
 - q : 55.825661
 - $\hat{q}^\beta$: 8.867082
 - q^β : 11.171213
 - Breakpoints: [0, 8.941176470587976, 181.68117647058827]
 - Error: 2.304130747483091
 - Relative error: 0.206256097388660
- (4, 3, 2)
 - q : 23.668856
 - $\hat{q}^\beta$: 6.406120
 - q^β : 6.675847
 - Breakpoints: [0, 19.92063386984238, 181.14352941176475]
 - Error: 0.269727110241886
 - Relative error: 0.040403430560780
- (5, 1, 1)

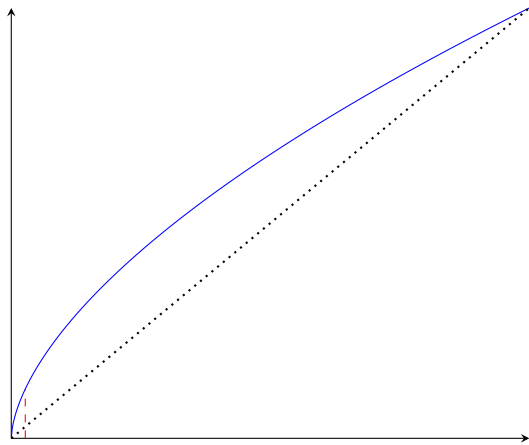
- q : 1.465633
- $\hat{q}^\beta$: 0.242922
- q^β : 1.257810
- Breakpoints: [0, 89.41176470588236]
- Error: 1.014888110054148
- Relative error: 0.806868876301326
- (5, 4, 2)
 - q : 11.133626
 - $\hat{q}^\beta$: 2.302152
 - q^β : 4.246019
 - Breakpoints: [0, 51.43476571725509, 192.3352941176471]
 - Error: 1.943867806538739
 - Relative error: 0.457809454712022
- $cu, 3$
 - q : 16.738997
 - $\hat{q}^\beta$: 4.837398
 - q^β : 5.422976
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 128.44338870924773]
 - Error: 0.585577497639370
 - Relative error: 0.107980844365705
- $cu, 5$
 - q : 20.944873
 - $\hat{q}^\beta$: 5.753021
 - q^β : 6.203641
 - Breakpoints: [0, 9.832103793393394, 14.682070969056184, 127.1036185995295]
 - Error: 0.450619295135236
 - Relative error: 0.072637877357868



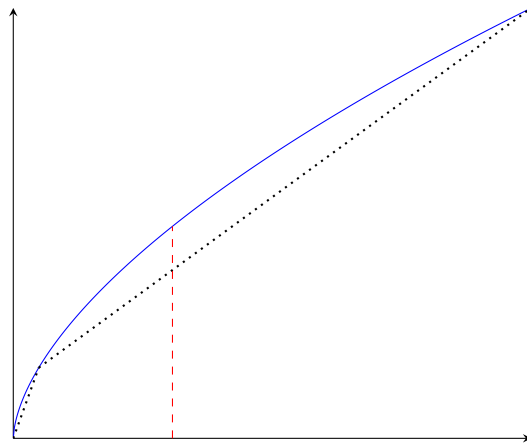
(a) $q^\beta - (1, 1, 2)$



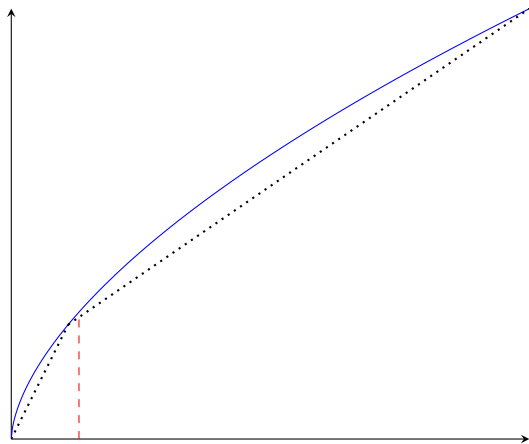
(b) $q^\beta - (2, 2, 1)$



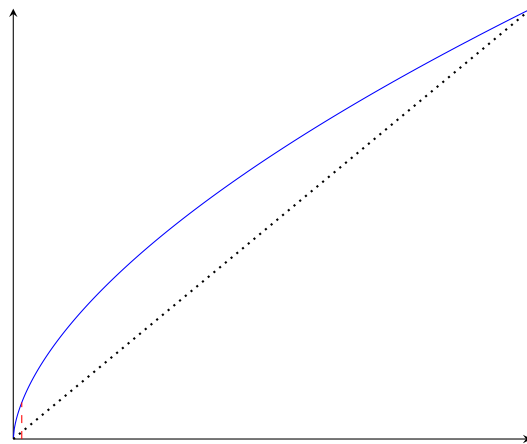
(c) $q^\beta - (2, 4, 2)$



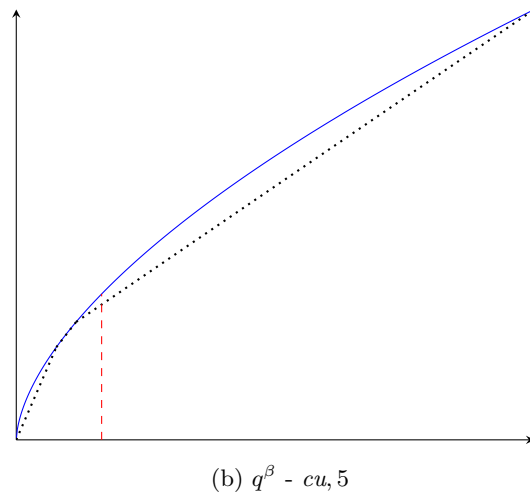
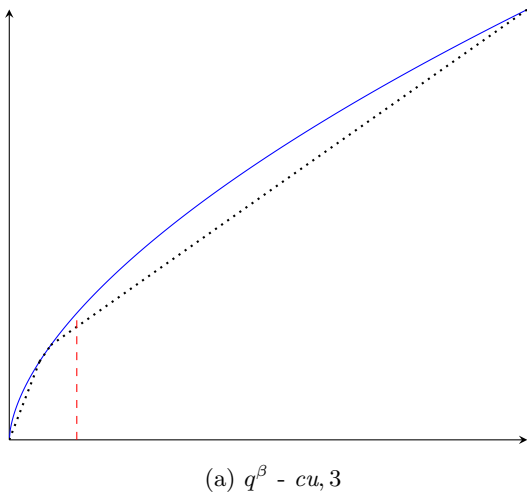
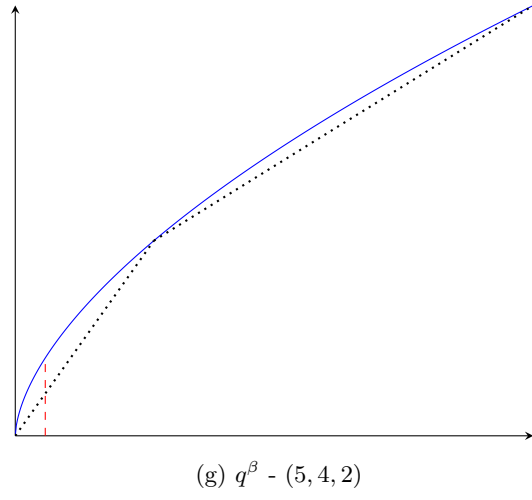
(d) $q^\beta - (3, 5, 2)$



(e) $q^\beta - (4, 3, 2)$



(f) $q^\beta - (5, 1, 1)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	-0.0	$z_{4,5,1}$	0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{4,5,2}$	0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	0.0	$z_{5,1,2}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,2,2}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,2}$	-0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,1}$	0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,1}$	0.0	$z_{5,5,2}$	-0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,2}$	0.0	$z_{cu,1}$	0.0	$A_{3,5,1}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{cu,2}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,4}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{hu,1}$	0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,2}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,3}$	0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,4}$	0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	0.0	$z_{hu,5}$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,1}$	0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,4,2}$	0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{3,5,1}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,1,2}$	0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,1,2}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,2,2}$	0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,3,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{2,1,1}$	111.1
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,2}$	27.842452830186858
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,1,3}$	100.0
$A_{1,2,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,2,3}$	22.2
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,2}$	50.04245283019103
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,3,3}$	100.0
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,5,2}$	50.042452830190236
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,5,3}$	44.5
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,1,2}$	0.0	$A_{cu,1}$	0.0	$dt_{3,1,2}$	54.4
$A_{2,2,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,3,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,4,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{3,1,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,2}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	50.0
$A_{3,2,1}$	0.0	$dt_{1,2,1}$	17.15006133296989	$dt_{3,4,2}$	37.7
$A_{3,2,2}$	0.0	$dt_{1,2,2}$	17.15006133296989	$dt_{3,4,3}$	21.1
$A_{3,3,1}$	0.0	$dt_{1,2,3}$	10.0	$dt_{3,5,1}$	22.3
$A_{3,3,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,3,2}$	66.7	$dt_{4,1,2}$	10.0
$A_{3,4,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,1,3}$	88.9
$A_{3,5,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,4,2}$	10.0	$dt_{4,2,2}$	155.5
$A_{4,1,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,2,3}$	33.3
$A_{4,2,1}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	10.0	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	9.414928732183045
$dt_{4,4,3}$	66.7	$f_{2,1,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,1}$	122.2	$f_{2,2,2}^H$	5.825472547254724	$f_{1,2,1}^C$	0.0
$dt_{4,5,2}$	10.0	$f_{2,3,1}^H$	4.774527452745195	$f_{1,2,2}^C$	0.0
$dt_{4,5,3}$	55.6	$f_{2,3,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,1,3}$	32.2	$f_{2,4,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348401	$f_{2,5,1}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,2}$	10.0	$f_{2,5,2}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,1}^H$	14.8	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,1,2}^H$	0.0	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,1}^H$	0.0	$f_{2,1,1}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,2,2}^H$	0.0	$f_{2,1,2}^C$	1.7303999999999995
$dt_{5,4,1}$	22.2	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,5,1}$	105.6	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,4,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{cu,1}$	53.3	$f_{3,5,1}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	12.6	$f_{2,5,2}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{3,1,1}^C$	5.120776699029128
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	8.4
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	17.3
$f_{1,2,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,1,2}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,2}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,1}^H$	0.0	$f_{4,3,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,1}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,4,2}^C$	0.0	$RecLMTD_{3,3,2}$	0.026450800778310225	$q_{1,3,2}$	0.0
$f_{4,5,1}^C$	13.9	$RecLMTD_{3,4,1}$	0.018541908861448915	$q_{1,4,1}$	0.0
$f_{4,5,2}^C$	0.0	$RecLMTD_{3,4,2}$	0.023514283865328507	$q_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$RecLMTD_{3,5,1}$	0.1	$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$RecLMTD_{4,1,1}$	0.019775319883283904	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$RecLMTD_{4,1,2}$	0.02150212335787724	$q_{2,1,1}$	0.0
$f_{5,3,1}^C$	0.0	$RecLMTD_{4,2,1}$	0.027922618193149045	$q_{2,1,2}$	0.0
$f_{5,3,2}^C$	0.0	$RecLMTD_{4,2,2}$	0.029297885186065308	$q_{2,2,2}$	0.0
$f_{5,4,1}^C$	0.0	$RecLMTD_{4,3,1}$	0.1	$q_{2,3,1}$	0.0
$f_{5,5,1}^C$	0.0	$RecLMTD_{4,4,1}$	0.02230651890829312	$q_{2,3,2}$	0.0
$f_{5,5,2}^C$	0.0	$RecLMTD_{4,4,2}$	0.024030191698075007	$q_{2,4,1}$	0.0
$RecLMTD_{1,1,1}$	0.1	$RecLMTD_{4,5,1}$	0.024784155348990776	$q_{2,5,1}$	0.0
$RecLMTD_{1,2,1}$	0.1	$RecLMTD_{4,5,2}$	0.02554670259920889	$q_{2,5,2}$	0.0
$RecLMTD_{1,2,2}$	0.1	$RecLMTD_{5,1,2}$	0.028630492980339938	$q_{3,1,1}$	0.0
$RecLMTD_{1,3,1}$	0.1	$RecLMTD_{5,2,1}$	0.1	$q_{3,1,2}$	0.0
$RecLMTD_{1,3,2}$	0.03070641897615819	$RecLMTD_{5,2,2}$	0.1	$q_{3,2,1}$	0.0
$RecLMTD_{1,4,1}$	0.1	$RecLMTD_{5,3,1}$	0.1	$q_{3,2,2}$	0.0
$RecLMTD_{1,4,2}$	0.1	$RecLMTD_{5,3,2}$	0.02888821823214753	$q_{3,3,1}$	0.0
$RecLMTD_{1,5,1}$	0.1	$RecLMTD_{5,4,1}$	0.1	$q_{3,3,2}$	0.0
$RecLMTD_{1,5,2}$	0.1	$RecLMTD_{5,5,1}$	0.1	$q_{3,4,1}$	0.0
$RecLMTD_{2,1,1}$	0.1	$RecLMTD_{5,5,2}$	0.1	$q_{3,4,2}$	0.0
$RecLMTD_{2,1,2}$	0.01500430180748921	$RecLMTD_{cu,1}$	0.016108908592626696	$q_{3,5,1}$	0.0
$RecLMTD_{2,2,2}$	0.1	$RecLMTD_{cu,2}$	0.009423141115769086	$q_{4,1,1}$	0.0
$RecLMTD_{2,3,1}$	0.024670522525974917	$RecLMTD_{cu,4}$	0.008535278616151328	$q_{4,1,2}$	0.0
$RecLMTD_{2,3,2}$	0.020926522199679467	$RecLMTD_{hu,1}$	0.012387051800747543	$q_{4,2,1}$	0.0
$RecLMTD_{2,4,1}$	0.025165980106347276	$RecLMTD_{hu,2}$	0.04561830526586298	$q_{4,2,2}$	0.0
$RecLMTD_{2,5,1}$	0.02833130321344378	$RecLMTD_{hu,3}$	0.06244696905967838	$q_{4,3,1}$	0.0
$RecLMTD_{2,5,2}$	0.1	$RecLMTD_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$RecLMTD_{3,1,1}$	0.014880108791979249	$RecLMTD_{hu,5}$	0.024141568940966463	$q_{4,4,2}$	0.0
$RecLMTD_{3,1,2}$	0.1	$q_{1,1,1}$	0.0	$q_{4,5,1}$	0.0
$RecLMTD_{3,2,1}$	0.1	$q_{1,2,1}$	0.0	$q_{4,5,2}$	0.0
$RecLMTD_{3,2,2}$	0.1	$q_{1,2,2}$	0.0	$q_{5,1,2}$	0.0
$RecLMTD_{3,3,1}$	0.1	$q_{1,3,1}$	0.0	$q_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,1}^H$	226.7	$t_{2,1,1}^C$	129.7960526315789
$q_{5,3,1}$	0.0	$t_{3,2,2}^H$	145.46969816954646	$t_{2,1,2}^C$	60.0
$q_{5,3,2}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,4,1}$	0.0	$t_{3,3,2}^H$	145.46969816954646	$t_{2,3,1}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,2}$	0.0	$t_{3,4,2}^H$	145.46969816954646	$t_{2,4,1}^C$	82.2
$q_{cu,1}$	0.0	$t_{3,5,1}^H$	226.7	$t_{2,5,1}^C$	204.4
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,2}^C$	93.3
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	271.1	$t_{3,1,1}^C$	127.37864077669906
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,2}^C$	60.0
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,2,1}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	221.1
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,3,2}^C$	37.8
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	176.7
$t_{1,2,1}^H$	160.0	$t_{4,5,2}^H$	271.1	$t_{3,4,2}^C$	82.2
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	126.72824858757069	$t_{3,5,1}^C$	204.4
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	60.0
$t_{1,3,2}^H$	93.3	$t_{5,2,2}^H$	65.6	$t_{4,1,2}^C$	60.0
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	115.6
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,2,2}^C$	115.6
$t_{1,5,1}^H$	160.0	$t_{5,4,1}^H$	198.9	$t_{4,3,1}^C$	37.8
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	82.2
$t_{2,1,1}^H$	248.9	$t_{5,5,2}^H$	126.72824858757063	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	137.8	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,1}^C$	204.4
$t_{2,2,2}^H$	137.8	$t_{1,2,1}^C$	115.6	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,3,2}^H$	137.8	$t_{1,3,1}^C$	221.1	$t_{5,2,1}^C$	115.6
$t_{2,4,1}^H$	248.9	$t_{1,3,2}^C$	37.8	$t_{5,2,2}^C$	115.6
$t_{2,5,1}^H$	248.9	$t_{1,4,1}^C$	102.85953757225435	$t_{5,3,1}^C$	221.1
$t_{2,5,2}^H$	137.8	$t_{1,4,2}^C$	176.7	$t_{5,3,2}^C$	37.8
$t_{3,1,1}^H$	226.7	$t_{1,5,1}^C$	93.3	$t_{5,4,1}^C$	156.04046242774564
$t_{3,1,2}^H$	65.6	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	93.3

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	93.3	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,1}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,1}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	1699.5893248312077	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	802.750117011701	$b_{1,2,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	1188.3798829882792	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	3355.16	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	0.0	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	1699.5893248312077
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	802.750117011701	$b_{1,2,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	1188.3798829882792	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0
$b_{3,5,1}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,5,2}^{H,out}$	0.0	$b_{1,4,2}^{C,in}$	0.0
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	3355.16	$b_{1,5,1}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	0.0	$b_{1,5,2}^{C,in}$	0.0
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	0.0	$b_{2,1,1}^{C,in}$	0.0
$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	103.82399999999997
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	705.16

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	819.3242718446605	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,1}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	103.82399999999997		
$b_{3,3,1}^{C,in}$	1857.24	$b_{2,2,2}^{C,out}$	705.16		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	3056.91	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,1}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	819.3242718446605		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	1857.24		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	2841.1600000000003	$b_{3,4,1}^{C,out}$	3056.91		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,4,1}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	2841.1600000000003		
$b_{1,2,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0		